2

Continuous Parameter Stochastic Process Theory

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2.1 Spatial Stochastic Processes

In this chapter, we consider probability models for a spatial variable that varies over a continuous domain of interest, $\mathcal{D} \subseteq \mathbb{R}^d$, where the spatial dimension is typically $d = 2$ or $d = 3$. Our approach relies on the notion of a spatial stochastic process $\{Y(\boldsymbol{s}) : \boldsymbol{s} \in \mathcal{D} \subseteq \mathbb{R}^d\}$, in the sense that

$$Y(\boldsymbol{s}) = Y(\boldsymbol{s}, \omega) \tag{2.1}$$

spatial location $\boldsymbol{s} \in \mathcal{D} \subseteq \mathbb{R}^d$ chance $\omega \in \Omega$

is a collection of random variables with a well-defined joint distribution. At any single spatial location $\boldsymbol{s} \in \mathcal{D}$, we think of $Y(\boldsymbol{s})$ as a random variable that can more fully be written as $Y(\boldsymbol{s}; \omega)$, where the elementary event ω lies in some abstract sample space, Ω. If we restrict attention to any fixed, finite set of spatial locations $\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n\} \subset \mathcal{D}$, then

$$(Y(\boldsymbol{s}_1), \ldots, Y(\boldsymbol{s}_n))^T \tag{2.2}$$

is a random vector, whose multivariate distribution reflects the spatial dependencies in the variable of interest. Each component corresponds to a spatial site. Conversely, if we fix any elementary event $\omega \in \Omega$, then

$$\{Y(\boldsymbol{s}, \omega) : \boldsymbol{s} \in \mathcal{D} \subseteq \mathbb{R}^d\} \quad \text{and} \quad (y_1, \ldots, y_n)^T = (Y(\boldsymbol{s}_1, \omega), \ldots, Y(\boldsymbol{s}_n, \omega))^T$$

are realizations of the spatial stochastic process (2.1) and the induced random vector (2.2), respectively. The observed data are considered but one such realization. A generalization

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to be discussed in Chapter 28 is that of a multivariate spatial stochastic process, for which $Y(\boldsymbol{s})$ is a random vector rather than just a random variable.

In applications, the sample space remains abstract, and the dependency on elementary events is suppressed in the notation. However, it is important to ensure a valid mathematical specification of the spatial stochastic process. Specifically, the distribution of the process $\{Y(\boldsymbol{s}) : \boldsymbol{s} \in \mathcal{D} \subseteq \mathbb{R}^d\}$ is given by the associated collection of the finite-dimensional joint distributions

$$F(y_1, \ldots, y_n; \boldsymbol{s}_1, \ldots, \boldsymbol{s}_n) = \mathbb{P}(Y(\boldsymbol{s}_1) \leq y_1, \ldots, Y(\boldsymbol{s}_n) \leq y_n) \tag{2.3}$$

of the random vector (2.2) for every n and every collection $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n$ of sites in $\mathcal{D}$. The celebrated Kolmogorov existence theorem states that the stochastic process model is valid if the family of the finite-dimensional joint distributions is consistent under reordering of the sites and marginalization. Intuitively, this is unsurprising; however, the details are cumbersome and we refer to Billingsley (1986) for a technical exposition.

An important special case is that of a *Gaussian* process where the finite-dimensional distributions (2.3) are multivariate normal and, therefore, characterized by their mean vectors and covariance matrices. In this particular case, the consistency conditions of the Kolmogorov existence theorem reduce to the usual requirement that covariance matrices are nonnegative definite. The case of non-Gaussian spatial stochastic processes is considerably more complex, and we refer the reader to Chapter 11.

2.2 Stationary and Intrinsically Stationary Processes

A spatial stochastic process is called *strictly stationary* if the finite dimensional joint distributions are invariant under spatial shifts. Essentially, this means that for all vectors $\boldsymbol{h} \in \mathbb{R}^d$ we have

$$F(y_1, \ldots, y_n; \boldsymbol{s}_1 + \boldsymbol{h}, \ldots, \boldsymbol{s}_n + \boldsymbol{h}) = F(y_1, \ldots, y_n; \boldsymbol{s}_1, \ldots, \boldsymbol{s}_n).$$

In the case of a Gaussian process $\{Y(\boldsymbol{s}) : \boldsymbol{s} \in \mathbb{R}^d\}$, where the finite dimensional distributions are determined by their second-order properties, we get, in particular, that

$$\mathbb{E}(Y(\boldsymbol{s})) = \mathbb{E}(Y(\boldsymbol{s} + \boldsymbol{h})) = \mu$$

and

$$\mathrm{Cov}(Y(\boldsymbol{s}), Y(\boldsymbol{s} + \boldsymbol{h})) = \mathrm{Cov}(Y(\boldsymbol{0}), Y(\boldsymbol{h})) = C(\boldsymbol{h}),$$

where the function $C(\boldsymbol{h}), \boldsymbol{h} \in \mathbb{R}^d$, is called the *covariance function*. A process, be it Gaussian or not, which satisfies these two conditions is called *weakly stationary* or *second-order stationary*. It follows that a Gaussian process, which is second-order stationary, is also strictly stationary. This is a very special property that depends critically on the Gaussian assumption.

Matheron (1971) proposed the use of the *semivariogram* or *variogram*, γ, which he defined as

$$\gamma(\boldsymbol{h}) = \frac{1}{2}\,\mathrm{var}(Y(\boldsymbol{s} + \boldsymbol{h}) - Y(\boldsymbol{s})),$$

as an alternative to the covariance function. Elementary calculations show that if Y is a second-order stationary spatial stochastic process with covariance function $C(\boldsymbol{h})$, then

$$\gamma(\boldsymbol{h}) = C(\boldsymbol{0}) - C(\boldsymbol{h}).$$

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The variogram can be used in some cases where a covariance function does not exist. More generally, Matheron (1973) introduced the class of the *intrinsically stationary* processes, which are such that certain spatial increments are second-order stationary, so that a *generalized covariance function* can be defined. For details, we refer to Chapter 11, Cressie (1993, Sec. 5.4), and Chilès and Delfiner (1999, Chap. 4).

Classical *Brownian motion* in one dimension provides an example of a Gaussian process that is intrinsically stationary, but not stationary. This process has variogram $\gamma(h) = |h|$, stationary and independent increments, and continuous sample paths. Interesting generalizations include Brownian motion and the *Brownian sheet* on spatial domains (Khoshnevisan, 2002). The Brownian motion process in one dimension allows for a series expansion in terms of a certain orthonormal system, which is often referred to as the *reproducing kernel Hilbert space* for the nonstationary Brownian motion covariance, $\mathrm{Cov}(Y(s), Y(t)) = \min(s, t)$ for $s, t > 0$. The coefficients in the expansion are independent, identically distributed, Gaussian random variables. This representation is commonly called a *Karhunen–Loève expansion* and relates closely to the Fourier representation discussed in Chapter 5. It can be generalized to much broader classes of spatial processes, under regularity conditions that generally correspond to the continuity of sample paths. See Breiman (1968, Chap. 12.7) and Wahba (1990) for details.

2.3 Nugget Effect

The classical geostatistical model of Chapter 3 decomposes a spatial stochastic process as

$$Y(\boldsymbol{s}) = \mu(\boldsymbol{s}) + \eta(\boldsymbol{s}) + \epsilon(\boldsymbol{s})$$

where $\mu(\boldsymbol{s}) = \mathbb{E}(Y(\boldsymbol{s}))$, the mean function, is deterministic and smooth, the process $\eta(\boldsymbol{s})$ has mean zero and continuous realizations, $\epsilon(\boldsymbol{s})$ is a field of spatially uncorrelated mean zero errors, and the processes η and ϵ are independent. The error process ϵ has covariance function

$$\mathrm{Cov}(\epsilon(\boldsymbol{s}), \epsilon(\boldsymbol{s}+\boldsymbol{h})) = \begin{cases} \sigma^2 \geq 0, & \boldsymbol{h} = \boldsymbol{0}, \\ 0, & \boldsymbol{h} \neq \boldsymbol{0}, \end{cases} \tag{2.4}$$

and is often referred to as a *nugget effect*. The nugget effect describes the observational error in (potentially) repeated measurements at any single site, or to microscale variability, occurring at such small scales that is cannot be distinguished from the effect of measurement error. The terminology stems from mining applications, where the occurrence of gold nuggets shows substantial microscale variability. Figure 2.1 shows a realization from a second-order stationary spatial stochastic process in the Euclidean plane with and without a nugget component. Without the nugget effect, the sample path is smooth; with the nugget effect, it is irregular and nondifferentiable.

In the following, we restrict our attention to the continuous part, $\eta(\boldsymbol{s})$, and to second-order stationary processes. Hence, from now on we consider spatial stochastic processes, which admit a continuous covariance function. The nugget effect (2.4) is the only discontinuous covariance function of practical interest (Gneiting, Sasvári and Schlather, 2001).

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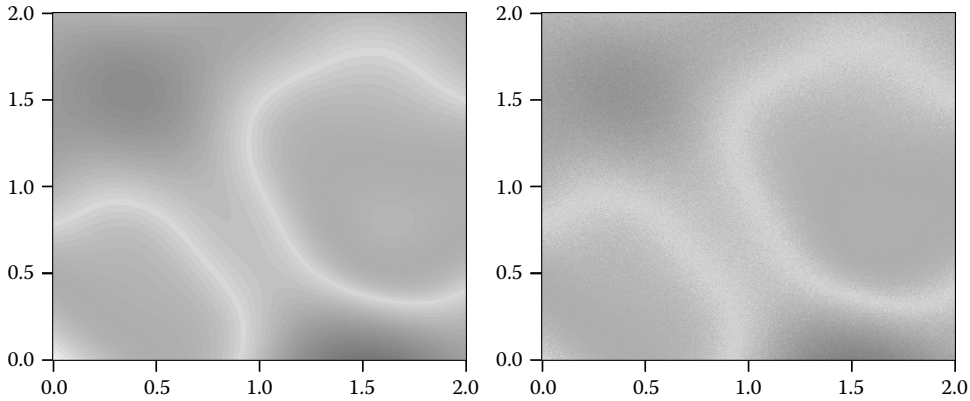


FIGURE 2.1
Realizations of a planar Gaussian process with isotropic Matérn covariance function (2.13) with smoothness parameter $\nu = \frac{3}{2}$ and scale parameter $\theta = 1$, without (left) and with a (very small) nugget effect (right). The sample paths were generated using the R package RANDOMFIELDS (Schlather, 2001).

2.4 Bochner's Theorem

Suppose that $\{Y(\boldsymbol{s}) : \boldsymbol{s} \in \mathbb{R}^d\}$ is a second-order stationary spatial stochastic process with covariance function C. Given any finite set of spatial locations $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n \in \mathbb{R}^d$, the covariance matrix of the finite dimensional joint distribution (2.3) is

$$\begin{pmatrix} C(\boldsymbol{0}) & C(\boldsymbol{s}_1 - \boldsymbol{s}_2) & \cdots & C(\boldsymbol{s}_1 - \boldsymbol{s}_n) \\ C(\boldsymbol{s}_2 - \boldsymbol{s}_1) & C(\boldsymbol{0}) & \cdots & C(\boldsymbol{s}_2 - \boldsymbol{s}_n) \\ \vdots & \vdots & \ddots & \vdots \\ C(\boldsymbol{s}_n - \boldsymbol{s}_1) & C(\boldsymbol{s}_n - \boldsymbol{s}_2) & \cdots & C(\boldsymbol{0}) \end{pmatrix}, \tag{2.5}$$

which needs to be a valid (that is, a nonnegative definite) covariance matrix. Another way of expressing this fact is to state that the covariance function is *positive definite*.* Conversely, given any positive definite function C that generates valid covariance matrices via (2.5) there exists a spatial Gaussian process with covariance function C.

By a classical theorem of Bochner (1933, 1955), a real-valued continuous function C is positive definite if and only if it is the Fourier transform of a symmetric, nonnegative measure F on $\mathbb{R}^d$, that is, if and only if

$$C(\boldsymbol{h}) = \int_{\mathbb{R}^d} \exp(i\, \boldsymbol{h}^T \boldsymbol{x})\, \mathrm{d}F(\boldsymbol{x}) = \int_{\mathbb{R}^d} \cos(\boldsymbol{h}^T \boldsymbol{x})\, \mathrm{d}F(\boldsymbol{x}). \tag{2.6}$$

We refer to (2.6) as the *spectral representation* of the covariance function. In particular, any real-valued characteristic function can serve as a valid correlation function. In most cases of practical interest, the *spectral measure*, F, has a Lebesgue density, f, called the *spectral density*, so that

$$C(\boldsymbol{h}) = \int_{\mathbb{R}^d} \exp(i\, \boldsymbol{h}^T \boldsymbol{x})\, f(\boldsymbol{x})\, \mathrm{d}\boldsymbol{x} = \int_{\mathbb{R}^d} \cos(\boldsymbol{h}^T \boldsymbol{x})\, f(\boldsymbol{x})\, \mathrm{d}\boldsymbol{x}. \tag{2.7}$$

* A function C on $\mathbb{R}^d$ is *positive definite* if the matrix (2.5) is nonnegative definite for all finite collections of sites $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n \in \mathbb{R}^d$. It is *strictly positive definite* if the matrix (2.5) is positive definite for all collections of distinct sites $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n \in \mathbb{R}^d$. The terminology used here, which differs for matrices and functions, stems from an unfortunate, yet well established, tradition in the literature.

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If the covariance function, C, is integrable over $\mathbb{R}^d$, the spectral density and the covariance are related via the standard Fourier inversion formula,

$$f(\boldsymbol{x}) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \cos(\boldsymbol{h}^T\boldsymbol{x})\, C(\boldsymbol{h})\, \mathrm{d}\boldsymbol{h}.$$

Chapter 5 provides a much more detailed discussion of spectral representations for stochastic processes and spectral methods in spatial statistics. Spectral densities also play a critical role in asymptotic theory for inference in spatial Gaussian processes (see Chapter 6 and Stein (1999)).

2.5 Isotropic Covariance Functions

A particular case of second-order stationary processes is when the covariance function, $C(\boldsymbol{h})$, depends on the spatial separation vector, $\boldsymbol{h}$, only through its Euclidean length, $\|\boldsymbol{h}\|$. We then call both the process and the covariance function *isotropic*. Note that the isocovariance curves for an isotropic process are circles or spheres around the point with which we are computing the covariance. A slight extension is that to *geometrically anisotropic* processes for which the isocovariance curves are ellipsoids rather than circles or spheres. See Chapter 3 for details. In applications, the isotropy assumption is frequently violated; however, isotropic processes remain fundamental, in that they form the basic building blocks of more complex, anisotropic and nonstationary spatial stochastic process models. Guan, Sherman and Calvin (2004) and Fuentes (2005), among others, propose tests for stationarity and isotropy in spatial data.

Without loss of generality, we may assume a standardized process, so that the covariance function satisfies $C(\mathbf{0}) = 1$. For an isotropic covariance, we can then write

$$C(\boldsymbol{h}) = \varphi(\|\boldsymbol{h}\|), \qquad \boldsymbol{h} \in \mathbb{R}^d, \tag{2.8}$$

for some continuous function $\varphi : [0, \infty) \to \mathbb{R}$ with $\varphi(0) = 1$. Let Φ_d denote the class of the continuous functions φ that generate a valid isotropic covariance function in $\mathbb{R}^d$ via the relationship (2.8). It is then obvious that

$$\Phi_1 \supseteq \Phi_2 \supseteq \cdots \qquad \text{and} \qquad \Phi_d \downarrow \Phi_\infty = \textstyle\bigcap_{d\geq 1} \Phi_d,$$

because we can restrict an isotropic process in $\mathbb{R}^d$ to any lower-dimensional subspace. However, an element of the class Φ_d need not belong to $\Phi_{d'}$ if $d' > d$. Armstrong and Jabin (1981) give a simple, striking example, in that an isotropic, triangular-shaped correlation function is valid in one dimension, but not in higher dimensions. The members of the class Φ_∞ are valid in all dimensions.

Schoenberg (1938) studied Bochner's representation in the special case of an isotropic or spherically symmetric function. He showed that a function $\varphi : [0, \infty) \to \mathbb{R}$ belongs to the class Φ_d if and only if it is of the form

$$\varphi(t) = \int_{[0,\infty)} \Omega_d(rt)\, \mathrm{d}F_0(r), \tag{2.9}$$

where F_0 is a probability measure on the positive half-axis, often referred to as the *radial spectral measure*, and

$$\Omega_d(t) = \Gamma(d/2) \left(\frac{2}{t}\right)^{(d-2)/2} J_{(d-2)/2}(t), \tag{2.10}$$

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TABLE 2.1

Generator Ω_d of the Class Φ_d

Dimension d	**1**	**2**	**3**	$\cdots$	∞
$\Omega_d(t)$	$\cos t$	$J_0(t)$	$t^{-1}\sin t$	$\cdots$	$\exp(-t^2)$
Lower bound for $\Omega_d(t)$	-1	-0.403	-0.218	$\cdots$	0

where Γ is Euler's gamma function and J is a Bessel function. In other words, the members of the class Φ_d are scale mixtures of a *generator*, Ω_d, and as such they have lower bound $\inf_{t\geq 0} \Omega_d(t)$. Table 2.1 and Figure 2.2 provide closed-form expressions, numerical values for the bound, and a graphical illustration. For example, in dimension $d = 2$, the generator is $\Omega_2(t) = J_0(t)$ and the lower bound is $\inf_{t\geq 0} J_0(t) = -0.403$.

In most cases of practical interest, the radial spectral measure, F_0, is absolutely continuous with *radial spectral density*, f_0, which relates to the spectral density, f, in Bochner's representation (2.7) through the relationship

$$f_0(\|\boldsymbol{x}\|) = 2\,\frac{\pi^{d/2}}{\Gamma(d/2)}\,\|x\|^{d-1} f(x), \qquad x \in \mathbb{R}^d.$$

Matheron (1965, 1973) and Gneiting (2002) describe operators, such as the *turning bands operator*, the *descente* and the *montée*, that build on spectral representations to map elements of the class Φ_d to some other class $\Phi_{d'}$.

Schoenberg (1938) furthermore showed that a function $\varphi : [0, \infty) \to \mathbb{R}$ belongs to the class Φ_∞ if and only if it is of the form

$$\varphi(t) = \int_{[0,\infty)} \exp(-r^2 t^2)\, \mathrm{d}F(r), \tag{2.11}$$

where F is a probability measure on $[0, \infty)$. One way of seeing this is by noting that $\lim_{d\to\infty} \Omega_d((2d)^{1/2}\, t) = \exp(-t^2)$ uniformly in $t \geq 0$. The members of the class Φ_∞ therefore are scale mixtures of a squared exponential generator, $\Omega_\infty(t) = \exp(-t^2)$. In particular, they are strictly positive and strictly decreasing functions, and have infinitely many derivatives away from the origin.

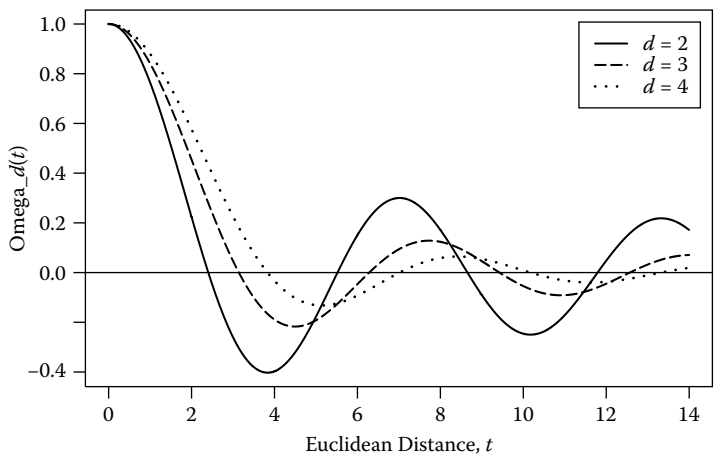


FIGURE 2.2
Generator Ω_d of the class Φ_d.

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In general, it can be difficult to check whether or not a function φ belongs to the class Φ_d and generates a valid isotropic covariance function. Hence, it is advisable to work with a parametric family that is known to be valid. From these known examples, many others can be constructed because sums, convex mixtures, products, and convolutions of valid covariance functions remain valid, whether or not the components are isotropic. In dimension $d = 1$, Pólya's criterion gives a simple sufficient condition: If φ is continuous and convex with $\varphi(0) = 1$ and $\lim_{t\to\infty} \varphi(t) = 0$, then $\varphi \in \Phi_1$. For example, the criterion applies if $\varphi(t) = \exp(-t)$ or $\varphi(t) = (1 + t)^{-1}$. Gneiting (2001) describes similar criteria in dimension $d \geq 2$.

2.6 Smoothness Properties

A spatial stochastic process $\{Y(\boldsymbol{s}) : \boldsymbol{s} \in D \subseteq \mathbb{R}^d\}$ is called *mean square continuous* if $\mathbb{E}(Y(\boldsymbol{s}) - Y(\boldsymbol{s} + \boldsymbol{h}))^2 \to 0$ as $\|\boldsymbol{h}\| \to 0$. For a second-order stationary process,

$$\mathbb{E}(Y(\boldsymbol{s}) - Y(\boldsymbol{s} + \boldsymbol{h}))^2 = 2(C(\boldsymbol{0}) - C(\boldsymbol{h})),$$

and, thus, mean square continuity is equivalent to the covariance function being continuous at the origin (and therefore everywhere). However, a process that is mean square continuous need not have continuous sample paths and vice versa (Banerjee and Gelfand, 2003). Adler (1981, Sec. 3.4) gives a covariance condition that guarantees the existence of a version with continuous sample paths for a stationary Gaussian process, and Kent (1989) has similar criteria that apply to any, Gaussian or non-Gaussian, stationary random field.

Turning now to isotropic processes, the properties of the member φ of the class Φ_d translate into properties of the associated Gaussian spatial process on $\mathbb{R}^d$ (Cramér and Leadbetter, 1967; Adler, 1981; Banerjee and Gelfand, 2003). In particular, the behavior of $\varphi(t)$ at the origin determines the smoothness of the sample paths, which is of great importance in spatial prediction problems (Stein, 1999). Specifically, suppose that there exists an $\alpha \in (0, 2]$ such that

$$1 - \varphi(t) \sim t^{\alpha} \quad \text{as} \quad t \downarrow 0, \tag{2.12}$$

then the realizations of the associated Gaussian spatial process have fractal or Hausdorff dimension $D = d + 1 - \frac{\alpha}{2}$. The larger α, the smaller the dimension D, and the smoother the realizations. If $\alpha = 2$, then the symmetrically continued function $c(u) = \varphi(|u|)$, $u \in \mathbb{R}$, is at least twice differentiable and the following holds for all positive integers m: The function $c(u)$ is $2m$ times differentiable at the origin if and only if the sample, paths of the associated Gaussian spatial process admit m derivatives. For instance, Figure 2.3 shows sample paths of Gaussian stochastic processes where $\varphi(t) = \exp(-t)$ and $\varphi(t) = (1 + t)\exp(-t)$, that is, with the Matérn covariance (2.13) with smoothness parameter $\nu = \frac{1}{2}$ and $\nu = \frac{3}{2}$. In the former case, $\alpha = 1$, so the realization is not differentiable; in the latter $\alpha = 2$ and the sample path admits $m = 1$ derivative. Note that these are properties of the stochastic process model; when plotting sample paths, visual appearance depends on the resolution, which needs to be chosen judiciously.

2.7 Examples of Isotropic Covariance Functions

We now give examples of parametric families within the class Φ_d that generate valid isotropic covariance functions via the relationship (2.8).

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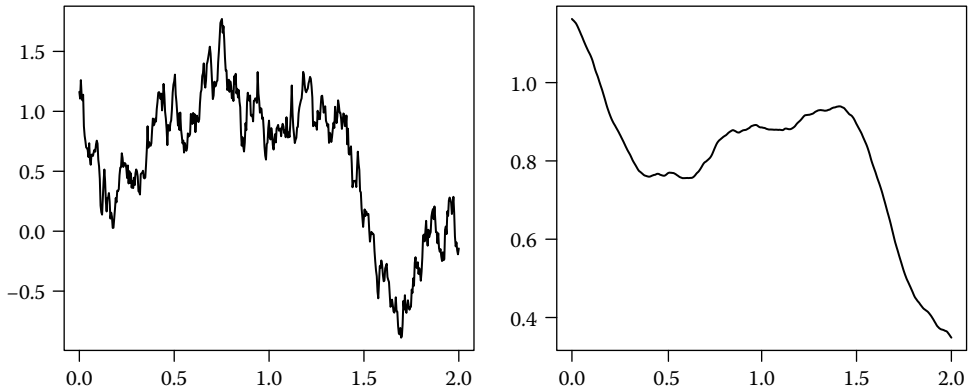


FIGURE 2.3
Sample paths of the one-dimensional Gaussian process with Matérn correlation function (2.13) where $\theta = 1$. The smoothness parameter is $\nu = \frac{1}{2}$ (left) and $\nu = \frac{3}{2}$ (right). For both panels, the resolution on the horizontal axis is in increments of 0.004.

The most popular and most often used family is the *Matérn class* (Matérn, 1960; Handcock and Stein, 1993; Guttorp and Gneiting, 2006), for which

$$\varphi(t) = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{t}{\theta}\right)^{\nu} K_{\nu}\left(\frac{t}{\theta}\right) \qquad (\nu > 0,\ \theta > 0), \tag{2.13}$$

where K_ν is a modified Bessel function, and $\nu > 0$ and $\theta > 0$ are smoothness and scale parameters, respectively. The Matérn correlation function (2.13) admits the relationship (2.12) where $\alpha = 2\min(\nu, 1)$. The associated Gaussian sample paths are m times differentiable if and only if $m < \nu$, as illustrated in Figure 2.3.

The members of the Matérn family belong to the class Φ_∞, so they are valid in all dimensions $d \geq 1$, with a spectral density function that is proportional to $(\theta^2\|\boldsymbol{x}\|^2 + 1)^{-\nu-\frac{d}{2}}$. Some special cases are noted in Table 2.2 and illustrated in Figure 2.4. If $\nu = \frac{1}{2}$, we obtain the exponential correlation function, $\varphi(t) = \exp(-t)$. The nugget effect (2.4) arises in the limit as $\nu \to 0$. The squared exponential correlation function, $\varphi(t) = \exp(-t^2)$, is also a limiting member, arising in the limit as $\nu \to \infty$ with the scale parameter set at $\theta = 1/(2\sqrt{\nu})$.

Whittle (1954, 1963) studied the stochastic fractional differential equation

$$\left(\frac{\partial^2}{\partial s_1^2} + \cdots + \frac{\partial^2}{\partial s_d^2} - \frac{1}{\theta^2}\right)^{(2\nu+d)/4} Y(\boldsymbol{s}) = \epsilon(\boldsymbol{s}),$$

TABLE 2.2
Special Cases of the Matérn Correlation Function (2.13) with Scale Parameter $\theta = 1$

Parameter	Correlation Function
$\nu = \frac{1}{2}$	$\varphi(t) = \exp(-t)$
$\nu = 1$	$\varphi(t) = t\,K_1(t)$
$\nu = \frac{3}{2}$	$\varphi(t) = (1+t)\,\exp(-t)$
$\nu = \frac{5}{2}$	$\varphi(t) = (1 + t + \frac{t^2}{3})\exp(-t)$

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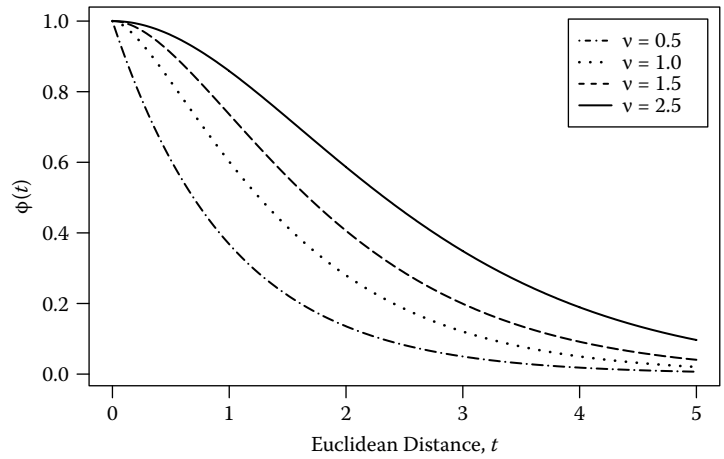


FIGURE 2.4
Special cases of the Matérn correlation function (2.13) with scale parameter $\theta = 1$.

where $\boldsymbol{s} = (s_1, \ldots, s_d)'$ and ϵ is a spatial white noise process. Essentially, the latter is a Gaussian process such that integrals of it become a spatial Brownian motion. Whittle (1963) showed that the solution to the stochastic differential equation is a Gaussian process with isotropic Matérn correlation function of the form (2.13). This construction can be thought of as an analog of the autoregressive moving average (ARMA) approach to time series modeling (Box and Jenkins, 1970) in continuous space. For a similar approach to lattice processes, see Chapter 13.

Another important class of correlation functions is the *powered exponential family*, which was employed by Diggle, Tawn and Moyeed (1998), among others. Its members are of the form

$$\varphi(t) = \exp\left(-\left(\frac{t}{\theta}\right)^{\alpha}\right) \qquad (0 < \alpha \leq 2,\ \theta > 0), \tag{2.14}$$

belong to the class Φ_∞, and admit the relationship (2.12).[†] Hence, the smoothness parameter $\alpha \in (0, 2]$ determines the fractal dimension of the Gaussian sample paths, while $\theta > 0$ is a scale parameter. However, the sample paths are either infinitely differentiable (when $\alpha = 2$) or not differentiable at all (when $\alpha < 2$), so the powered exponential family provides a less flexible parametrization than the Matérn class. The exponential correlation function arises when $\alpha = 1$.

Gneiting and Schlather (2004) introduced the *Cauchy family*, whose members are of the form

$$\varphi(t) = \left(1 + \left(\frac{t}{\theta}\right)^{\alpha}\right)^{-\beta/\alpha} \qquad (0 < \alpha \leq 2,\ \beta > 0,\ \theta > 0) \tag{2.15}$$

and belong to the class Φ_∞. The smoothness parameter $\alpha \in (0, 2]$ and the scale parameter $\theta > 0$ are interpreted in the same way as for the exponential power family. However, there is an additional long-memory parameter, $\beta > 0$, that determines the asymptotic power law, $\varphi(t) \sim t^{-\beta}$, with which the correlation decays as $t \to \infty$. The smaller the β, the stronger the long-range dependence (Beran, 1994). Long-memory dependence is unlikely to be relevant in spatial interpolation, but can be of critical importance in problems of estimation and inference for spatial data (Whittle, 1962; Haslett and Raftery, 1989).

[†] If $\alpha > 2$, the powered exponential model (2.14) does not belong to any of the classes Φ_d. Similarly, if $\alpha > 2$ the Cauchy model (2.15) does not belong to any of the classes Φ_d.

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The Matérn, powered exponential and Cauchy families belong to the class Φ_∞, that is, they generate valid isotropic covariance functions in all spatial dimensions. Consequently, they admit the representation (2.11) and so they are strictly positive and strictly decreasing.

In applications, these assumptions might be too restrictive. For example, moderate negative correlations are occasionally observed and referred to as a *hole effect*. Figure 2.2 shows that the generator Ω_d of the class Φ_d exhibits this type of behavior. Another option is to fit an *exponentially damped cosine* function,

$$\varphi(t) = \exp\left(-\tau\frac{t}{\theta}\right)\cos\left(\frac{t}{\theta}\right) \qquad \left(\tau \geq \frac{1}{\tan\frac{\pi}{2d}},\ \theta > 0\right), \tag{2.16}$$

where the restriction on the decay parameter, τ, is necessary and sufficient for membership in the class Φ_d (Zastavnyi, 1993).

Covariance functions with *compact support* allow for computationally efficient estimation, prediction, and simulation (Gneiting, 2002; Furrer, Genton and Nychka, 2006). The most popular, compactly supported model is the *spherical* correlation function with scale parameter $\theta > 0$, for which

$$\varphi(t) = 1 - \frac{3}{2}\frac{t}{\theta} + \frac{1}{2}\left(\frac{t}{\theta}\right)^3 \tag{2.17}$$

if $t < \theta$ and $\varphi(t) = 0$ otherwise. This function belongs to the class Φ_d if $d \leq 3$, but is not valid in higher dimensions. Many other models with compact support are available (Gneiting, 2002). For example, the compactly supported function

$$\varphi(t) = \left(1 + \frac{80}{47}\frac{t}{\theta} + \frac{2500}{2209}\frac{t^2}{\theta^2} + \frac{32000}{103823}\frac{t^3}{\theta^3}\right)\left(1 - \frac{10}{47}\frac{t}{\theta}\right)^8 \tag{2.18}$$

if $t < \frac{47}{10}\theta$ and $\varphi(t) = 0$ otherwise, where $\theta > 0$ is a scale parameter, is valid in dimension $d \leq 3$, is smooth, and approximates the squared exponential correlation function, in that $\sup_{t\geq 0}|\varphi(t) - \exp(-t^2/(2\theta^2))| = 0.0056$.

2.8 Prediction Theory for Second-Order Stationary Processes

The most common problem in spatial statistics is to predict, or interpolate, the value of the process at a location $\boldsymbol{s} \in \mathbb{R}^d$ where no observation has been made. If we want to do least squares prediction, that is, if we wish to minimize the expected squared prediction error, it is well known (Ferguson, 1967) that the optimal predictor $\hat{Y}(\boldsymbol{s})$ is the conditional expectation given the observations, that is,

$$\hat{Y}(\boldsymbol{s}) = \mathbb{E}\left(Y(\boldsymbol{s}) \,|\, Y(\boldsymbol{s}_1) = y_1, \ldots, Y(\boldsymbol{s}_n) = y_n\right). \tag{2.19}$$

Generally speaking, this predictor is difficult to evaluate in closed form. However, the Gaussian case is an exception, for if

$$\begin{pmatrix} U \\ V \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_U \\ \mu_V \end{pmatrix}, \begin{pmatrix} \Sigma_{UU} & \Sigma_{UV} \\ \Sigma_{VU} & \Sigma_{VV} \end{pmatrix}\right),$$

then the conditional distribution of V given U is normal,

$$(V|U) \sim \mathcal{N}\left(\mu_V + \Sigma_{VU}\Sigma_{UU}^{-1}(U - \mu_U),\ \Sigma_{VV} - \Sigma_{VU}\Sigma_{UU}^{-1}\Sigma_{UV}\right). \tag{2.20}$$

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Let $V = Y(\boldsymbol{s})$, and let $U = (Y(\boldsymbol{s}_1), \ldots, Y(\boldsymbol{s}_n))^T$ denote the observed data. Assuming that we have a second-order stationary process with mean μ and covariance function C, so that $\mu_U = \mu \mathbf{1}^T \in \mathbb{R}^n$, $\mu_V = \mu$, $\Sigma_{UU} = [C(\boldsymbol{s}_i - \boldsymbol{s}_j)]_{i,j=1}^n \in \mathbb{R}^{n \times n}$, $\Sigma_{UV} = [C(\boldsymbol{s} - \boldsymbol{s}_i)]_{i=1}^n \in \mathbb{R}^n$ and $\Sigma_{VV} = C(\mathbf{0})$, Equation (2.19) and Equation (2.20) give

$$\hat{Y}(\boldsymbol{s}) = \mu + (C(\boldsymbol{s} - \boldsymbol{s}_1), \ldots, C(\boldsymbol{s} - \boldsymbol{s}_n))[C(\boldsymbol{s}_i - \boldsymbol{s}_j)]^{-1} \begin{pmatrix} Y(\boldsymbol{s}_1) - \mu \\ \vdots \\ Y(\boldsymbol{s}_n) - \mu \end{pmatrix}. \tag{2.21}$$

We see that for a Gaussian process with a known mean and a known covariance function the conditional expectation is linear in the observed data. Moreover, the *ordinary kriging predictor* defined by (2.21) is the best linear predictor in the least squares sense even when the process is not necessarily Gaussian (Whittle, 1963, Chap. 5; Ripley, 1981, Sec. 4.4). If one estimates the covariance, the predictor is no longer linear and, thus, not optimal. For additional discussion, see Chapter 3.

By elementary matrix manipulations, the ordinary kriging predictor (2.21) can be rewritten in *dual kriging* form, that is,

$$\hat{Y}(\boldsymbol{s}) = \mu + (Y(\boldsymbol{s}_1) - \mu, \ldots, Y(\boldsymbol{s}_n) - \mu) \left[C(\boldsymbol{s}_i - \boldsymbol{s}_j)\right]^{-1} \begin{pmatrix} C(\boldsymbol{s} - \boldsymbol{s}_1) \\ \vdots \\ C(\boldsymbol{s} - \boldsymbol{s}_n) \end{pmatrix}. \tag{2.22}$$

From this we see that, when viewed as a function of $\boldsymbol{s} \in \mathbb{R}^d$, the ordinary kriging predictor is a linear combination of a constant and terms of the form $w_i\, C(\boldsymbol{s} - \boldsymbol{s}_i)$, where the coefficients, w_i, depend on the observed data and the (known) second-order structure, akin to interpolation with radial basis functions (Wendland, 2005; Fasshauer, 2007). In other words, the predicted surface interpolates the observations (unless there is a nugget effect) and inherits its appearance from the covariance function, in that locally, in the neighborhood of an observation site, $\boldsymbol{s}_i$, it behaves like the covariance function at the origin.

It is important to note that the kriging predictor may no longer be optimal if we abandon the quadratic loss function that underlies least squares prediction. Under other loss functions, such as linear or piecewise linear loss, other predictors might be optimal. For example, the conditional median is optimal if we wish to minimize the expected absolute error, rather than the expected squared error (Ferguson, 1967), a distinction that can become important in non-Gaussian settings.

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